

## PAKSE, Lao People's Democratic Republic

WMO: 489550

Lat: 15.136N

Lon: 105.779E

Elev: 102

StdP: 100.11

Time Zone: 7.00 (E07)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | 16.3       | 17.7 | 10.1                          | 7.8 | 21.6 | 11.2 | 8.4 | 22.2 | 9.5                   | 24.4 | 7.8 | 22.8 | 2.0                      | 0    | 0.329 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 8.8                    | 36.7            | 25.5 | 35.7 | 25.3 | 34.9 | 25.1 | 27.5                | 32.7 | 27.1 | 32.2 | 26.8 | 31.7 | 1.8                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.2                            | 21.9 | 30.0 | 25.8 | 21.4 | 29.4 | 25.6 | 21.1 | 29.2 | 87.7          | 33.1 | 86.3 | 32.5 | 85.2 | 31.9 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 7.3           | 5.7  | 4.9 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 27.9   | 25.5  | 27.5  | 29.7  | 30.5  | 29.7  | 28.7  | 27.8  | 27.6  | 27.8  | 27.7  | 26.9  | 25.6  |
|                                                                        | DBStd       | 2.33   | 2.31  | 2.33  | 1.95  | 1.85  | 1.58  | 1.43  | 1.46  | 1.23  | 1.29  | 1.20  | 2.00  | 2.38  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD10.0     | 6537   | 481   | 491   | 611   | 615   | 610   | 559   | 552   | 547   | 533   | 547   | 507   | 485   |
|                                                                        | CDD18.3     | 3496   | 222   | 257   | 352   | 365   | 352   | 309   | 293   | 289   | 283   | 289   | 257   | 227   |
|                                                                        | CDH23.3     | 38757  | 2183  | 2946  | 4540  | 4817  | 4431  | 3575  | 3025  | 2918  | 2861  | 2876  | 2450  | 2136  |
| (14) Wind                                                              | WSAvg       | 1.9    | 2.0   | 2.0   | 2.2   | 1.9   | 1.7   | 1.7   | 1.7   | 1.6   | 1.6   | 1.7   | 2.2   | 2.2   |
|                                                                        | PrecAvg     | 2217   | 7     | 14    | 45    | 83    | 241   | 272   | 496   | 487   | 398   | 138   | 30    | 10    |
|                                                                        | PrecMax     | 2819   | 47    | 71    | 128   | 208   | 467   | 493   | 874   | 776   | 941   | 322   | 109   | 98    |
|                                                                        | PrecMin     | 1479   | 0     | 0     | 11    | 27    | 114   | 99    | 163   | 253   | 226   | 29    | 2     | 0     |
|                                                                        | PrecStd     | 356    | 12    | 17    | 30    | 49    | 86    | 110   | 172   | 146   | 163   | 73    | 25    | 19    |
|                                                                        | DB          | 34.4   | 36.0  | 37.1  | 38.8  | 37.3  | 35.0  | 34.2  | 33.2  | 33.7  | 34.1  | 34.4  | 34.6  | 34.6  |
|                                                                        | MCWB        | 22.9   | 24.0  | 25.0  | 25.9  | 26.6  | 26.7  | 26.2  | 26.9  | 26.9  | 25.7  | 25.3  | 23.9  | 23.9  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 33.4   | 34.9  | 36.1  | 37.3  | 35.6  | 33.7  | 32.8  | 32.1  | 32.5  | 33.0  | 33.4  | 33.6  | 33.6  |
|                                                                        | 2% MCWB     | 22.5   | 23.5  | 24.7  | 25.8  | 26.7  | 26.6  | 26.4  | 26.6  | 26.6  | 25.3  | 25.0  | 23.5  | 23.5  |
|                                                                        | 5% DB       | 32.5   | 33.9  | 35.2  | 36.0  | 34.5  | 32.8  | 31.8  | 31.2  | 31.7  | 32.2  | 32.4  | 32.4  | 32.4  |
|                                                                        | 5% MCWB     | 22.1   | 23.1  | 24.4  | 25.7  | 26.5  | 26.5  | 26.4  | 26.4  | 26.3  | 25.2  | 24.6  | 23.1  | 23.1  |
|                                                                        | 10% DB      | 31.3   | 32.9  | 34.4  | 34.9  | 33.5  | 31.9  | 30.8  | 30.4  | 30.8  | 31.3  | 31.4  | 31.1  | 31.1  |
|                                                                        | 10% MCWB    | 21.6   | 22.8  | 24.3  | 25.5  | 26.3  | 26.3  | 26.2  | 26.2  | 26.1  | 25.1  | 24.3  | 22.6  | 22.6  |
|                                                                        | WB          | 25.2   | 25.7  | 28.9  | 27.7  | 27.8  | 27.5  | 27.2  | 27.4  | 27.6  | 26.8  | 26.7  | 25.5  | 25.5  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | MCDB        | 30.3   | 33.6  | 34.2  | 35.2  | 33.8  | 32.7  | 31.5  | 31.5  | 31.9  | 31.5  | 31.8  | 30.9  | 30.9  |
|                                                                        | 2% WB       | 24.2   | 24.7  | 26.3  | 26.9  | 27.2  | 27.1  | 26.9  | 27.0  | 27.1  | 26.4  | 26.1  | 24.6  | 24.6  |
|                                                                        | 2% MCDB     | 29.8   | 32.6  | 33.2  | 34.1  | 33.2  | 32.0  | 31.0  | 30.9  | 31.2  | 30.8  | 31.4  | 30.6  | 30.6  |
|                                                                        | 5% WB       | 23.6   | 24.3  | 25.7  | 26.5  | 27.0  | 26.8  | 26.6  | 26.7  | 26.7  | 26.1  | 25.7  | 24.1  | 24.1  |
|                                                                        | 5% MCDB     | 29.6   | 31.7  | 32.9  | 33.6  | 32.9  | 31.4  | 30.5  | 30.3  | 30.6  | 30.2  | 30.6  | 30.2  | 30.2  |
|                                                                        | 10% WB      | 22.9   | 23.8  | 25.2  | 26.1  | 26.6  | 26.6  | 26.3  | 26.4  | 26.4  | 25.7  | 25.2  | 23.5  | 23.5  |
|                                                                        | 10% MCDB    | 29.3   | 30.8  | 32.2  | 33.1  | 32.2  | 30.9  | 29.9  | 29.7  | 30.0  | 29.6  | 29.5  | 29.5  | 29.5  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 12.0   | 11.3  | 10.0  | 8.8   | 7.6   | 6.1   | 5.5   | 5.4   | 5.8   | 7.5   | 9.0   | 10.9  | 10.9  |
|                                                                        | 5% DB       | 12.4   | 11.6  | 10.4  | 9.7   | 8.8   | 7.1   | 7.0   | 6.8   | 7.2   | 8.8   | 9.8   | 11.3  | 11.3  |
|                                                                        | MCWBR       | 3.8    | 3.6   | 3.3   | 2.7   | 2.9   | 2.4   | 2.5   | 2.6   | 2.7   | 3.2   | 3.3   | 3.6   | 3.6   |
|                                                                        | 5% WB       | 10.5   | 9.7   | 9.1   | 8.9   | 7.9   | 6.4   | 6.3   | 6.2   | 6.6   | 7.4   | 8.9   | 10.5  | 10.5  |
|                                                                        | MCWBR       | 3.8    | 3.6   | 3.6   | 3.1   | 2.8   | 2.5   | 2.6   | 2.7   | 2.8   | 2.9   | 3.4   | 3.9   | 3.9   |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.439  | 0.564 | 0.664 | 0.621 | 0.498 | 0.464 | 0.460 | 0.460 | 0.483 | 0.483 | 0.427 | 0.415 | 0.415 |
|                                                                        | taud        | 2.250  | 1.900 | 1.707 | 1.852 | 2.209 | 2.316 | 2.322 | 2.313 | 2.243 | 2.223 | 2.367 | 2.354 | 2.354 |
|                                                                        | Ebn at Noon | 851    | 758   | 696   | 727   | 810   | 830   | 835   | 844   | 827   | 818   | 858   | 864   | 864   |
|                                                                        | Edh at Noon | 134    | 198   | 245   | 212   | 146   | 130   | 130   | 132   | 142   | 141   | 118   | 118   | 118   |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 5.16   | 5.49  | 5.35  | 5.47  | 5.25  | 4.72  | 4.38  | 4.25  | 4.43  | 4.89  | 4.92  | 4.85  | 4.85  |
|                                                                        | RadStd      | 0.23   | 0.24  | 0.19  | 0.25  | 0.26  | 0.32  | 0.45  | 0.34  | 0.47  | 0.42  | 0.25  | 0.22  | 0.22  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | +0.16 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page